



HDB AWS DATA PLATFORM ARCHITECTURE

Target Architecture & ETL Pipeline Solution

Senior Data Engineer Assessment | Multi – VPC Architecture

Multi-VPC AWS Architecture Solves Ingestion, Security and Analytics Isolation

PUBLIC INGESTION ENGINE

Automated Batch Pull (>100MB)

- Event Bridge scheduled trigger invokes Lambda / Glue
- Fetches batch CSV/JSON datasets from data.gov.sg
- Uses NAT Gateway for controlled outbound internet access
- Writes ingestion data to S3 Raw via S3 VPC Endpoint
- Supports large files without Lambda memory buffering

ETL COMPUTE & 5 S3 TIERS

AWS Glue Spark & Data Quality

- PySpark performs profiling, validation and transformation
- Calculates remaining lease based on 99-year lease period
- SHA-256 hashes the unique Resale Identifier
- Segregates data into 5 governed S3 tiers:

Raw → Cleaned → Transformed → Hashed → Quarantined

ISOLATED ANALYTICS VPC

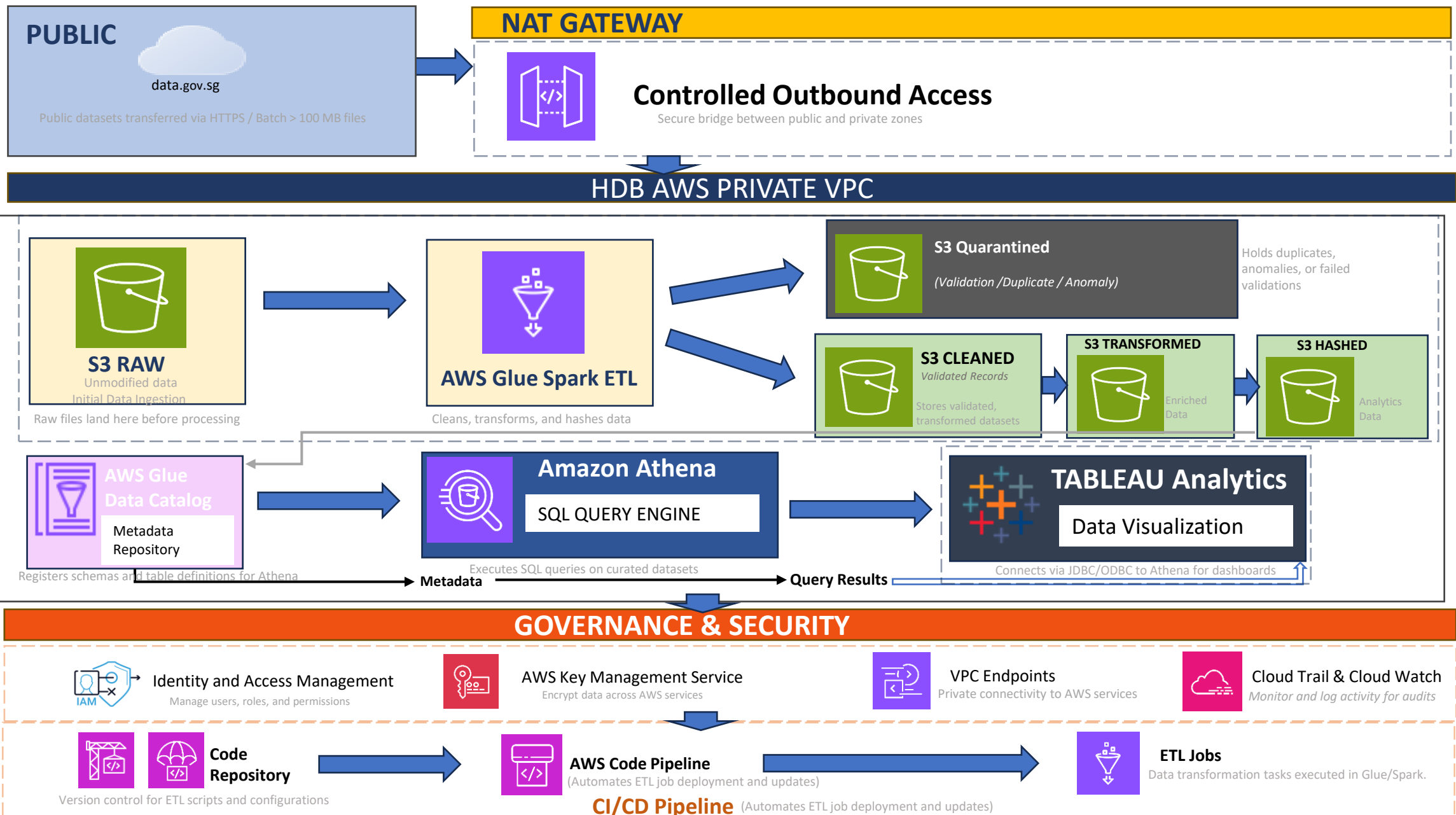
Tableau Cross-VPC Integration

- Tableau Server hosted in dedicated Data Science VPC
- Connects to Amazon Athena using JDBC/ODBC
- Private connectivity via AWS Private Link / VPC endpoints
- Athena queries the S3 Hashed layer without copying data
- Analytics environment has no direct public exposure

End-to-End Batch ETL Processes >100MB Files Across 5 Mandatory S3 Tiers

Output Tier	S3 Location Standard	Description & Data Quality	Consumer
1. Raw	s3://hdb-raw/	Unmodified historical batch files from data.gov.sg (Jan 2012 – Dec 2016). Preserves raw immutable audit lineage.	Glue Spark Ingestion Engine
2. Cleaned	s3://hdb-cleaned/	Validated records matching Jan 2012 authoritative set for Date, Town, Flat Type, Flat Model, and Storey Range.	Data Transformation Pipeline
3. Transformed	s3://hdb-transformed/	Enriched dataset with computed remaining lease (Years & Months) and composite key deduplication.	Downstream ETL
4. Quarantined	s3://hdb-quarantine/	Isolated records failing validation, composite key duplicates (lower price), and resale price anomalies.	Audit & Monitoring
5. Hashed	s3://hdb-hashed/	Final production dataset with irreversible SHA-256 hashed Resale Identifier preserving row uniqueness	Amazon Athena & Tableau

HDB Data Platform - AWS Solution Architecture



Step Functions, KMS, IAM, and CloudWatch Deliver Enterprise Governance

PIPELINE WORKFLOW

AWS Step Functions

Involves end-to-end ETL workflow.
Flow: Ingestion Trigger → Glue Spark ETL → Data Quality / Quarantine → Glue Crawler → Athena
Automated **retry and error-handling logic** improves pipeline resilience.
Workflow status and failures are centrally tracked.

DATA SECURITY & ENCRYPTION

AWS KMS Customer

Manage keys enforce SSE-KMS encryption across all 5 S3 tiers.
TLS 1.3 secures traffic between VPC subnets and endpoints.
Meets enterprise encryption and compliance standards.

IDENTITY & ACCESS MANAGEMENT

ACCESS CONTROL & PERMISSIONS

Strict least-privilege IAM roles for Glue, Lambda, and Athena.
Minimal S3 bucket policies applied.
Tableau restricted to Athena query execution only — no direct S3 access..

MONITORING & AUDIT

Amazon CloudWatch

Monitors ETL jobs, workflow status, errors and data-quality metrics.
AWS CloudTrail records AWS API activity for security and compliance auditing.
Centralized logging provides operational visibility, lineage and governance traceability.